



Codefez GCSE programming Project 2026-27

`codefez.github.io`

Scenario: Fez Flix Cinema Seating Reservation System

Background

Fez Flix Cinema is an exclusive cinema viewing experience. The one-screen cinema offers users a unique opportunity to watch the latest movies on uncomfortable, non-reclining seats reminiscent of the experience one would have endured had he or she visited a cinema in the 1950s. The experience has been gathering huge demand and screenings are often sold out.

Fez Flix requires a program to manage seat bookings for a single screening room, calculate ticket pricing, and generate exportable receipt files.

System Rules

1. **Seating Layout:** The screening room consists of 5 rows (labeled A to E) and 6 seats per row (labeled 1 to 6), represented using a 2D array.
2. **Seat Status:**
 - 0 represents an available seat.
 - X represents a reserved seat.
3. **Pricing Tiers:**
 - Standard Seats (Rows A–C): £8.00 each
 - VIP Seats (Rows D–E): £12.00 each
4. **Discounts:**
 - Student/Child ticket: 20% discount on the base seat price.
5. **Capacity Trigger:**
 - When occupancy reaches 70% capacity (21 of 30 seats booked), remaining available seats increase in price by a 15% surge factor.

Tasks

Task 1: Main Menu & Grid Display

Develop a main menu loop with the following options:

- Display Seating Chart
- Reserve Seats
- Display Booking Statistics
- Export Receipt File
- Quit

The seating chart must display clearly as a grid formatted with row letters (A–E) and column numbers (1–6).
See the diagram below

	1	2	3	4	5
A	O	X	O	O	O
B	O	X	X	X	O
C	O	X	O	X	X
D	O	X	X	O	O
E	O	X	O	O	O

Task 2: Seat Selection & Input Validation

Develop the reservation module to allow seat booking:

- Prompt the user for row letter and seat number.
- Validate that the row letter is between A–E and seat number is between 1–6.
- Validate that the selected seat is not already taken (X).
- Update the seating grid to mark successful bookings with an X.

Task 3: Ticket Pricing & Multi-Seat Purchases

Calculate costs based on seat tier and customer type:

- Ask whether each ticket is Adult or Student/Child, applying discounts where appropriate.
- Allow users to book multiple seats within a single transaction session and output a running total.

Task 4: File Export & Receipt Generation

Develop an export feature to save booking details:

- Prompt for customer name and generate a unique reference code.
- Check if `receipt.txt` already exists; ask whether to overwrite or prompt for a new filename.
- Write customer details, booked seat coordinates, individual ticket prices, and total bill to the text file.

Task 5: Dynamic Demand Surcharge Extension

Extend the program to track total occupancy:

- If total room bookings reach or exceed 21 seats (70% capacity), apply a 15% surge fee to all subsequent bookings.
- Display an explicit notification when high-demand pricing is active.
- Students still receive a 20% discount but the surcharge is still added to the ticket price after the student discount has been applied.

EXTENSION TASKS

- Research and use a Python module like `tabulate`, `pandas`, `prettytables` or `TKinter` to display your grid in a more aesthetic style.
- Create an option to select viewings on a schedule. For example over 7 days.
 - You can load the schedule from a file
 - And save changes back to a file
- **Multiple Screenings / Schedule Handling.** You can also create a schedule of 2-3 viewings a day
 - Expand the text file I/O to load an entire day's schedule (e.g., 3 screenings: 14:00, 17:30, 20:30).
 - Each screening should maintain its own independent 2D seating matrix.
- **Seat Cancellation & Modification:** Add a menu option allowing users to cancel a booking by row and seat number, updating the grid back to O and adjusting occupancy counts.
- **Block Booking / Adjacent Seats:** Allow users to request x seats together in a single row. The program must automatically search the 2D array and suggest consecutive available seats.
- **Revenue Analytics & Reporting:**
Add an administrative menu option (password-protected or hidden) that calculates total daily revenue, breakdown of ticket types sold (Adult vs. Student), and average seat occupancy percentage.
- **Extended User Interface & Aesthetics (GUI)**
- **Interactive Graphical UI (Tkinter / Pygame):**
 - Instead of console input, allow students to click seats on a graphical grid that toggle color (e.g., Green for available, Red for reserved) and dynamically update the price total on screen.